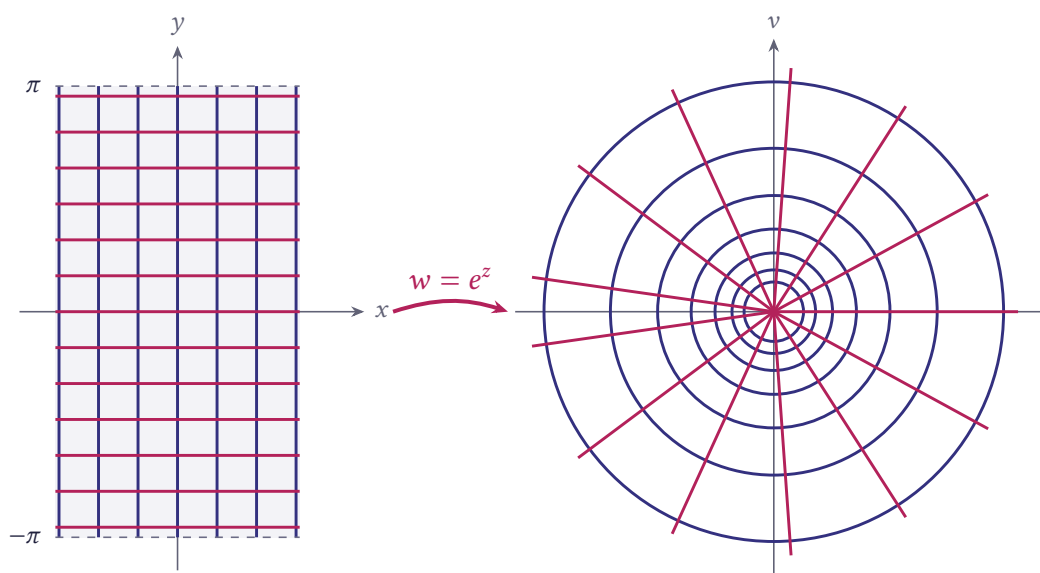


Elementary Complex Functions

توابع مختلط مقدماتی



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The elementary complex functions generalize the elementary real functions.

3.1 The Exponential Function

The real exponential function $f(x) = e^x$ has the following properties:

$$\begin{aligned} 1) \quad & f(x_1 + x_2) = f(x_1)f(x_2) & e^{x_1}e^{x_2} &= e^{x_1+x_2} \\ 2) \quad & \frac{d}{dx}f(x) = f(x) & \frac{d}{dx}e^x &= e^x \\ 3) \quad & e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \cdots \end{aligned}$$

We define the complex exponential function inspired by the third property, and then verify the other properties for it.

Definition 3.1 ■ The complex exponential function

$$w = e^z \triangleq 1 + z + \frac{z^2}{2!} + \cdots + \frac{z^n}{n!} + \cdots$$

If in the definition of the complex exponential we substitute $z = iy$:

$$\begin{aligned} e^{iy} &= 1 + iy + \frac{1}{2!}(iy)^2 + \frac{1}{3!}(iy)^3 + \cdots \\ &= 1 + iy - \frac{1}{2!}y^2 - \frac{i}{3!}y^3 + \frac{1}{4!}y^4 + \frac{i}{5!}y^5 + \cdots \\ &= \underbrace{\left(1 - \frac{1}{2!}y^2 + \frac{1}{4!}y^4 + \cdots\right)}_{\cos y} + i \underbrace{\left(y - \frac{1}{3!}y^3 + \frac{1}{5!}y^5 + \cdots\right)}_{\sin y} = \cos y + i \sin y \end{aligned}$$

Key result 3.2

$$w = e^z = e^{x+iy} = e^x e^{iy} = e^x (\cos y + i \sin y)$$

Example 3.3

Is the function $w = e^z$ analytic?

$$\begin{aligned} u(x, y) &= e^x \cos y, & v(x, y) &= e^x \sin y \\ \frac{\partial u}{\partial x} &= e^x \cos y = \frac{\partial v}{\partial y} \quad \checkmark & \frac{\partial u}{\partial y} &= -e^x \sin y = -\frac{\partial v}{\partial x} \quad \checkmark \end{aligned}$$

Since the Cauchy–Riemann conditions hold in the whole complex plane, the function $w = e^z$ is differentiable in the whole complex plane.

Example 3.4

Find the derivative of $w = e^z$.

$$\frac{d}{dz}e^z = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = e^x \cos y + ie^x \sin y = e^x(\cos y + i \sin y) = e^z$$

Key result 3.5 ■ Important remark

From now on we can write every complex number in exponential form:

$$z = x + iy = |z|(\cos \theta + i \sin \theta) = |z|e^{i\theta}$$

Observe that

$$e^{z+i2k\pi} = e^z e^{i2k\pi} = e^z [\cos(2k\pi) + i \sin(2k\pi)] = e^z$$

so $w = e^z$ is a periodic function in the complex plane with period $i2\pi$.

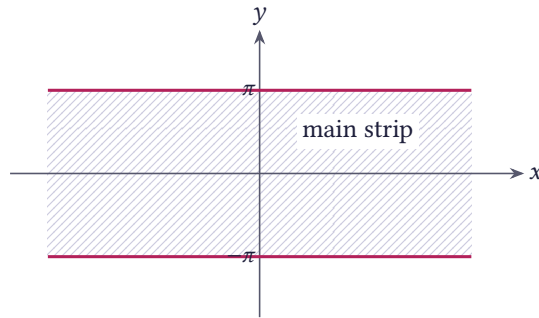


Figure 3.1 The fundamental strip of the exponential function: $-\pi < y \leq \pi$

3.2 Trigonometric Functions

$$\begin{aligned} e^{i\theta} &= \cos \theta + i \sin \theta \quad \implies \quad e^{-i\theta} = \cos \theta - i \sin \theta \\ \implies \quad \cos \theta &= \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} \end{aligned}$$

Definition 3.6 ■ Complex cosine and sine

$$\cos z \triangleq \frac{e^{iz} + e^{-iz}}{2}, \quad \sin z \triangleq \frac{e^{iz} - e^{-iz}}{2i}$$

$$\begin{aligned} \cos z &= \frac{1}{2} [e^{i(x+iy)} + e^{-i(x+iy)}] = \frac{1}{2} [e^{ix}e^{-y} + e^{-ix}e^y] \\ &= \frac{1}{2} [e^{-y}(\cos x + i \sin x) + e^y(\cos x - i \sin x)] \\ &= \cos x \left(\frac{e^y + e^{-y}}{2} \right) - i \sin x \left(\frac{e^y - e^{-y}}{2} \right) \\ &= \cos x \cosh y - i \sin x \sinh y \end{aligned}$$

$$\begin{aligned}
\sin z &= \frac{1}{2i} \left[e^{i(x+iy)} - e^{-i(x+iy)} \right] = \frac{1}{2i} \left[e^{ix} e^{-y} - e^{-ix} e^y \right] \\
&= \frac{1}{2i} \left[e^{-y} (\cos x + i \sin x) - e^y (\cos x - i \sin x) \right] \\
&= \sin x \left(\frac{e^y + e^{-y}}{2} \right) + i \cos x \left(\frac{e^y - e^{-y}}{2} \right) \\
&= \sin x \cosh y + i \cos x \sinh y
\end{aligned}$$

■ Analyticity and Differentiability of $\cos z$

$$\cos z = u(x, y) + iv(x, y) = \cos x \cosh y - i \sin x \sinh y, \quad u = \cos x \cosh y, \quad v = -\sin x \sinh y$$

$$\frac{\partial u}{\partial x} = -\sin x \cosh y = \frac{\partial v}{\partial y} \quad \checkmark \quad \frac{\partial u}{\partial y} = \cos x \sinh y = -\frac{\partial v}{\partial x} \quad \checkmark$$

So this function is analytic in the whole complex plane. We obtain its derivative with one of the methods described earlier:

$$f'(z) = \frac{d}{dz} \cos z = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = -\sin x \cosh y - i \cos x \sinh y = -\sin z$$

■ Analyticity and Differentiability of $\sin z$

$$\sin z = u(x, y) + iv(x, y) = \sin x \cosh y + i \cos x \sinh y, \quad u = \sin x \cosh y, \quad v = \cos x \sinh y$$

$$\frac{\partial u}{\partial x} = \cos x \cosh y = \frac{\partial v}{\partial y} \quad \checkmark \quad \frac{\partial u}{\partial y} = \sin x \sinh y = \frac{\partial v}{\partial x} \quad \checkmark$$

So this function is analytic in the whole complex plane and

$$f'(z) = \frac{d}{dz} \sin z = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \cos x \cosh y + i \sin x \sinh y = \cos z$$

Key result 3.7

$$(\sin z)' = \cos z, \quad (\cos z)' = -\sin z$$

■ Unboundedness of $\sin z$ and $\cos z$

$$|\cos z| = \sqrt{\cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y} = \sqrt{\cos^2 x + \sinh^2 y}, \quad (\cosh^2 y = 1 + \sinh^2 y, \quad \sin^2 x = 1 - \cos^2 x)$$

$$|\sin z| = \sqrt{\sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y} = \sqrt{\sin^2 x + \sinh^2 y}, \quad (\cosh^2 y = 1 + \sinh^2 y, \quad \cos^2 x = 1 - \sin^2 x)$$

Since $\sinh y$ is an unbounded function, the functions $\cos z$ and $\sin z$ are **unbounded**.

■ Periodicity

$$\begin{aligned}\cos(z + 2\pi) &= \cos(x + iy + 2\pi) = \cos(x + 2\pi) \cosh y - i \sin(x + 2\pi) \sinh y \\ &= \cos x \cosh y - i \sin x \sinh y = \cos z\end{aligned}$$

In the same way $\sin(z + 2\pi) = \sin z$.

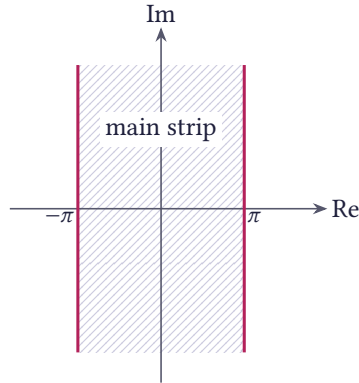


Figure 3.2 The fundamental strip of $\sin z$ and $\cos z$: $-\pi < x \leq \pi$

■ Remark

$$\cos(iy) = \cosh y, \quad \sin(iy) = i \sinh y$$

$$\cos(x + iy) = \cos x \cos(iy) - \sin x \sin(iy) = \cos x \cosh y - i \sin x \sinh y$$

$$\sin(x + iy) = \sin x \cos(iy) + \cos x \sin(iy) = \sin x \cosh y + i \cos x \sinh y$$

The other trigonometric functions:

$$\tan z \triangleq \frac{\sin z}{\cos z}, \quad \cot z \triangleq \frac{\cos z}{\sin z}, \quad \sec z \triangleq \frac{1}{\cos z}, \quad \csc z \triangleq \frac{1}{\sin z}$$

$$(\tan z)' = \sec^2 z, \quad \dots$$

3.3 Hyperbolic Functions

$$\cosh z \triangleq \frac{e^z + e^{-z}}{2}, \quad \sinh z \triangleq \frac{e^z - e^{-z}}{2}, \quad e^z = e^x(\cos y + i \sin y)$$

Exercise 3.8

Verify that

$$\cosh z = \cosh x \cos y + i \sinh x \sin y, \quad \sinh z = \sinh x \cos y + i \cosh x \sin y$$

Exercise 3.9

Verify that $\cosh z$ and $\sinh z$ are analytic in the whole complex plane and that their derivatives are given by

$$(\cosh z)' = \sinh z, \quad (\sinh z)' = \cosh z$$

$$\tanh z = \frac{\sinh z}{\cosh z}, \quad \coth z = \frac{\cosh z}{\sinh z}, \quad \operatorname{sech} z = \frac{1}{\cosh z}, \quad \operatorname{csch} z = \frac{1}{\sinh z}$$

$$\cosh(z + i2\pi) = \cosh(z), \quad \sinh(z + i2\pi) = \sinh(z)$$

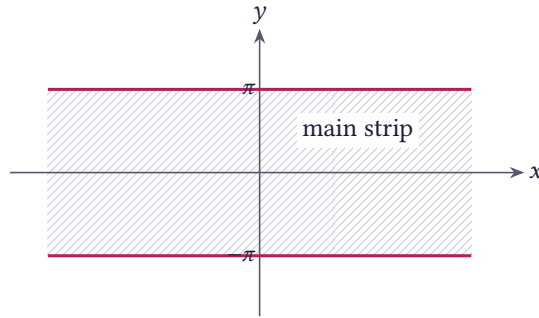


Figure 3.3 The fundamental strip of the hyperbolic functions: $-\pi < y \leq \pi$

3.4 The Logarithm Function

$$w = \ln z, \quad z = |z|e^{i \arg z} = e^w = e^{\ln |z|} e^{i \arg z}$$

$$w = \ln z = \ln |z| + i \arg z = \ln(x^2 + y^2)^{1/2} + i \tan^{-1} \frac{y}{x}$$

$$u(x, y) = \frac{1}{2} \ln(x^2 + y^2), \quad v(x, y) = \tan^{-1} \frac{y}{x}$$

$$\frac{\partial u}{\partial x} = \frac{x}{x^2 + y^2} = \frac{\partial v}{\partial y} \quad \checkmark \quad \frac{\partial u}{\partial y} = \frac{y}{x^2 + y^2} = -\frac{\partial v}{\partial x} \quad \checkmark$$

Hence the logarithm function is analytic in the whole complex plane except at the origin.

$$\frac{d}{dz} \ln z = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{x}{x^2 + y^2} + i \frac{-y}{x^2 + y^2} = \frac{\bar{z}}{z\bar{z}} = \frac{1}{z}$$

$$z = re^{i\theta} = re^{i(\theta + 2k\pi)}$$

So that the definition of a function is not violated, the range of this function is the region shown in the figure.

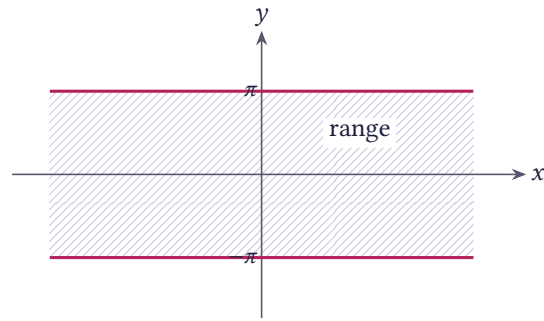


Figure 3.4 The range of the logarithm function: the strip $-\pi < \text{Im } w \leq \pi$

The following properties hold for the logarithm function:

$$\ln(z_1 \times z_2) = \ln z_1 + \ln z_2, \quad \ln\left(\frac{z_1}{z_2}\right) = \ln z_1 - \ln z_2, \quad \ln(z^c) = c \ln z$$